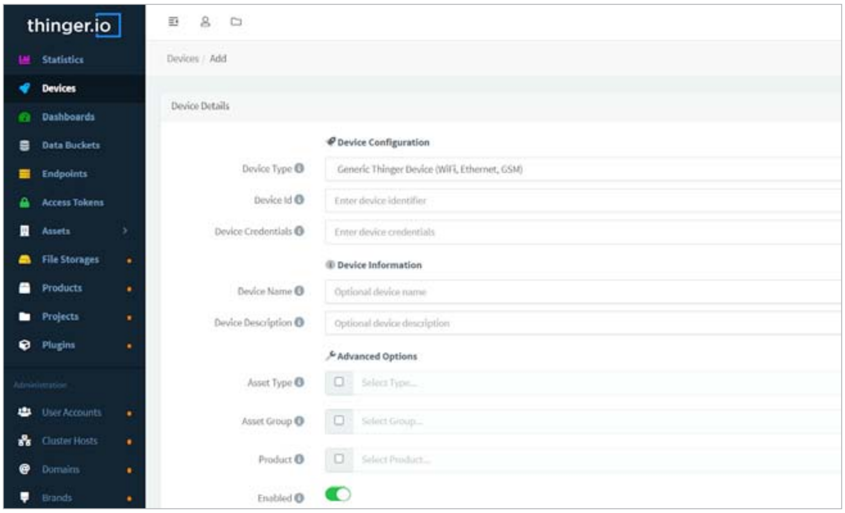

Figure 3: The *Device* tab from the menu

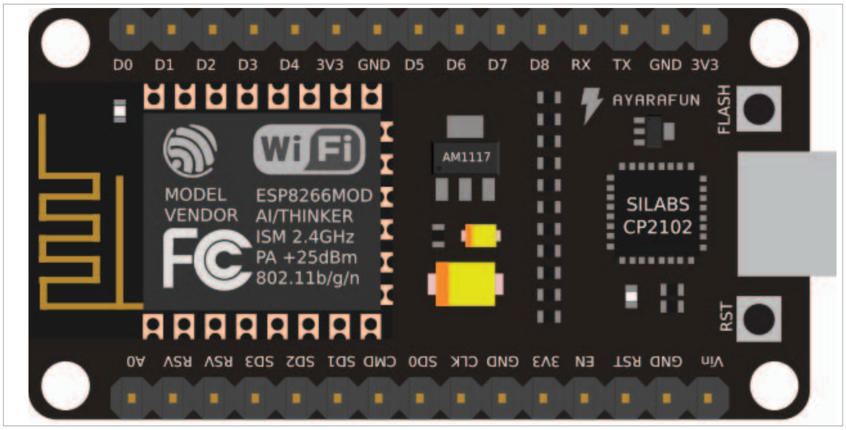

Figure 4: The ESP8266 chip

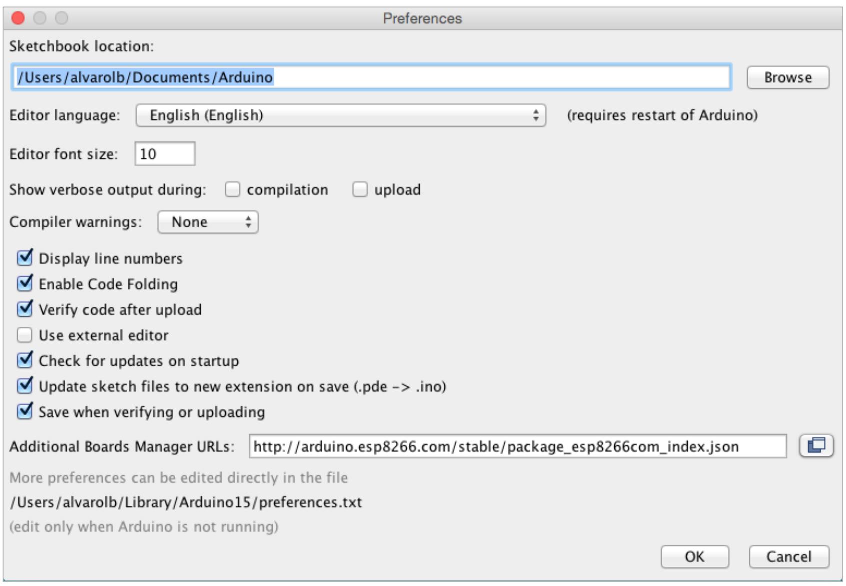

Figure 5: Adding the package for ESP8266 chip on Arduino IDE

> *Board manager*. Then search and install the esp8266 package (Figure 6).

This should now enable any program involving ESP8266 directly from the Arduino IDE. By going to *Tools* > *Boards* you should see the new ESP8266 boards installed. Select your board to be able to compile code for the ESP8266.

You can find additional information for the ESP8266 package in the ESP8266 GitHub repository. The easiest board to program is the NodeMCU, which does not require pressing *Flash* + *Reset* buttons for uploading the sketch. For other boards you will need to use a USB to serial converter (3v3!) and flash the sketch by setting some GPIOs to GND.

Use the code below to connect an ESP8266 device to the cloud platform in a few lines using the Wi-Fi interface. Please remember to modify the *arduino_secrets.h* file with your own information.

```
#define THINGER_SERIAL_DEBUG
#include <ThingerESP8266.h>
#include "arduino_secrets.h"
ThingerESP8266 thing(USERNAME, DEVICE_
ID, DEVICE_CREDENTIAL);

void setup() {
  // open serial for monitoring
  Serial.begin(115200);
  // set builtin led as output
  pinMode(LED_BUILTIN, OUTPUT);
  // add WiFi credentials
  thing.add_wifi(SSID, SSID_PASSWORD);
  // digital pin control example
(i.e. turning on/off a light, a relay,
configuring a parameter, etc)
  thing["led"] << digitalPin(LED_
BUILTIN);
  // resource output example (i.e.
reading a sensor value)
  thing["millis"] >>
outputValue(millis());
  // more details at http://docs.
thinger.io/arduino/
}
void loop() {
```